



ESPE

UNIVERSIDAD DE LAS FUERZAS ARMADAS

INNOVACIÓN PARA LA EXCELENCIA



Team 1:

Parking Lot System



Member:

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Object Oriented Programming

Feature 1 – Entry/Exit Management and Manual Space Allocation

Duration: 1 week

Goal: Ensure accurate control of vehicle entry and exit operations by allowing operators to manually register movements, validate license plates, and maintain real-time parking spot availability.

Task	Description
Time Stamp Verification	Ensure all entry and exit transactions are recorded with accurate date and time stamps for auditing and reporting purposes.
Operator Confirmation Step	Require manual confirmation from the parking operator for each entry/exit transaction to prevent errors and ensure oversight.
Duplicate Plate Detection	Implement a check to identify if a license plate is already recorded as present in the parking facility to prevent duplicate entries.
Spot Availability Validation	Verify and update real-time parking spot availability before allocating a space to an incoming vehicle.

Feature 2 – Resident Rental and Subscription Management

Duration: 2 week

Goal: Manage resident information, payments, and rent payment status to ensure authorized and up-to-date use of parking spaces.

Task	Description
Payment Date Recording	Log the exact date when a resident's rental or subscription payment is received.
Expiration Date Calculation	Automatically calculate the subscription or rental expiration date based on the payment date and plan duration.
Rental Status Validation	Check and display whether a resident's rental/subscription is active, expired, or pending renewal.
Subscription History Tracking	Maintain a log of all past payments, renewals, and plan changes for each resident.
Account Deactivation	Provide functionality to deactivate a resident's account and parking access upon request or due to non-payment.
Data Consistency Checks	Implement validation rules to ensure resident payment and status data remains consistent and error-free.
Rent renewal	Enable the process for residents to renew their parking rental or subscription, updating their status and expiration date.

Feature 3 -- Visitor and Resident Management

Duration: 1 week

Goal: It manages the registration of residents and visitors, their current status, and allows the updating of their personal data and the deletion of their data in the system.

Task	Description
Implement Resident Registration and Update Flow	Create forms and processes for adding new residents and editing their existing profile information.
Display Resident List with Basic Details	Screen that displays a list of all residents, including key information such as name, unit, and status.
Implement Visitor Registration Linked to Resident	Allow residents or operators to register visitors, linking them to the host resident's account.
Add Visitor Authorization Status Flag	Include a field to mark whether a visitor is pre-authorized by a resident or requires manual approval for entry.

Feature 4 -- Data Reporting and PDF Generation

Duration: 1 week

Goal: Generates reports of the most important data within the system, including resident and visitor reports, and vehicle entry and exit times. Allows download to your local computer in PDF format.

Task	Description
Implement General Vehicle List Table	Create a report table displaying all vehicles currently in the facility with details like license plate, type, and entry time.
Create Historical Entrances and Exits Log Table	Generate a record of all past vehicle entry and exit transactions for auditing purposes.
Implement Visitor Logs Table	Create a report table that shows all visitor entries and exits, including the host resident
Develop Screen-to-PDF Export Functionality	It allows users to export any report or data table directly to a PDF file for download.
Validate Data Accuracy in Generated Reports	Ensure all data presented in reports is consistent with the live database and correctly formatted for export.